

ECO 303—Intermediate Microeconomic Theory

Spring 2019

Department of Economics
Stony Brook University

Instructor: Alejandro Melo Ponce

Lecture time: Mondays and Wednesdays, 2:30PM–3:50PM

Classroom: Staller Center M0113

Office hours: Mondays and Wednesdays, 9:20AM–10:50AM SBS N604

E-mail: alejandro.meloponce@stonybrook.edu

Graduate TA: Jing Wei

Recitation: Mondays and Wednesdays, 4:00PM–4:53PM

Classroom: Melville Library N4072

Office hours: Mondays and Wednesdays, 5:00PM–6:00PM SBS N616

E-mail: jing.wei@stonybrook.edu

Undergraduate TAs: Kunyao Yan & Ningyu Ding

Office hours: Tuesday (Kunyao) 1:00PM–2:00PM, Thursday 4:00PM–5:00PM (Ningyu); SBS Economics Lobby

E-mail: kunyao.yan@stonybrook.edu, ningyu.ding@stonybrook.edu

Course Description

The purpose of this course is to introduce the foundational tools of microeconomic analysis. As you know from your Principles of Economics class, Economics is the study of choice, i.e. how individuals or groups make decisions in a world where resources are limited. The overarching topics in this course are how consumers make choices both under certainty and uncertainty; how firms take decisions subject to technological constraints and in relation to the market structure they operate in and the organization of markets in the presence of complete and asymmetric information. To this end, the course aims to provide students with the economic and analytical tools to understand these topics.

The real world is very complex, so in order to gain some understanding, economists construct simple models to examine difficult economic questions. In this course I will present some basic models and show how they can be used to understand real world applications. Thus, this class will focus on developing techniques for analyzing economic questions. Whereas in your Principles of Economics class the focus was on learning some facts about economics, now we will focus in developing the tools of modern microeconomics. By the end of the course, you will become comfortable in applying those analytical tools in answering economic questions and also in applying them in more advanced economics classes. In order to develop these tools we will use mathematical concepts to do it. Modern Microeconomics uses math in order to communicate with precision and rigor its results, so this course will be an introduction to how modern economics is conducted. That being said, it is not my intention to make this course artificially hard or use more math than is required. This is a course in economics that relies on math, not a math course with some economic applications. Modern economics requires some math, so we are going to use it when necessary. However, to keep the focus on economics, I will try to give intuition and to stress the economic significance of the tools and results that we will cover.

Course Requirements

ECO 303 requires a previous course in Principles of Economics, which in Stony Brook is ECO 108, with a grade of C or higher. This course also requires a previous course at the level of calculus or higher. At Stony Brook the minimum requirement is either MAT 125, AMS 151 with a grade of C or higher or placement level 5 in the mathematics placement examination. Mathematics is a very useful tool for reasoning with precision, so as detailed in the introduction, we will use it regularly.

More precisely, the mathematics requirement for this course is **single variable calculus**. One of the first topics that we will cover in class is to learn some elementary techniques from **multivariate calculus**, but the techniques are just an extension of the single variable case, so this is the reason why there is no formal prerequisite for a multivariate calculus class. Everything that we need with respect to new mathematical tools will be developed in class.

Course Material

- **Required.** *Intermediate Microeconomics: A Modern Approach* by Hal R. Varian, W. W. Norton. This is the required textbook for the course. The Fifth edition or above (currently in the 9th edition) is suitable for the course. The accompanying Workouts in Intermediate Microeconomics, also by Varian, might be useful for study purposes.
- **Recommended.** *Intermediate Microeconomics and its Application* by Walter Nicholson and Christopher Snyder, 12th edition, Cengage Learning, 2014. A more gentle treatment than the required text. Recommended as supplementary reading.
- **Recommended.** *Microeconomic Theory: Basic Principles and Extensions* by Walter Nicholson and Christopher Snyder, 11th edition, Cengage Learning, 2011. A more advanced treatment of Microeconomics for students that want to study the concepts in-depth and with a more rigorous and mathematical approach.
- **Recommended.** *Essential Mathematics for Economic Analysis* by K. Sydsaeter and P. Hammond, Prentice-Hall 2008. Additional reading covering the mathematical concepts that we will use.
- I will assign some readings to illustrate the models that we will cover in class and to provide real world illustrations.
- Any extra materials will be posted on BlackBoard so it is your responsibility to check it regularly.

Recitations

Each week there will be a recitation section conducted by the graduate TA Jing Wei. Each recitation will discuss past problems sets and elaborate and discuss the material from the lecture. Recitations sections will meet weekly beginning the second week of class. Students are strongly encouraged to attend their respective recitation sections because it provides an useful opportunity to work through problems, elaborate on concepts discussed in class and seek answers to questions.

Keeping up and course attendance

The course will move at a reasonable pace. Your priority every week should be to review the material of the week, including the lectures, the problem sets and any assigned reading. Material directly relevant to the exams but not in the textbook or in Blackboard may appear in the lectures. For the previous reason, attending the lectures and the recitations is strongly advised. By attending lectures you will be able to benefit from in-class presentations and discussions. You are also responsible for learning everything discussed in class whether these topics are in the textbook or not.

Rules and Regulations

During lectures, please keep your cell phones switched off and refrain from text messaging. Inattentive students whose actions are judged to be disruptive to class discussion will be asked to leave.

Grading Policy

Grading will be weighted as follows:

- Midterms (20%) each
- Problem Sets (25%)

- Final exam (35%)

A letter grade based on the sum of the components above will be assigned as follows:

Letter Grade	Score Achieved
A	[93, 100]
A–	[90, 93)
B+	[87, 90)
B	[83, 87)
B–	[80, 83)
C+	[77, 80)
C	[73, 77)
C–	[70, 73)
D+	[67, 70)
D	[60, 67)
F	[0, 60)

Problem Sets

We will have n Problem Sets during the course. I will announce in class and in BlackBoard when a new problem set is posted. Problems sets are due approximately **one week** after they are announced and you must deliver them at the end of the class. The graded problem sets will be returned in your recitation section, and any unclaimed problem sets will stay with the TA. The due date is strict. You are encouraged to discuss the problem sets in small groups of maximum five students. However, it is essential that you try to solve each question before you meet with your study group. It is indispensable that you attempt to solve the problems on your own if you wish to learn microeconomics. Each student will have to acknowledge the group members at the top of the solution to his/her problem set.

The problem sets can be downloaded from BlackBoard on the date they are announced. I will post answers on the day the problem set is due. Thus, late homework will not be accepted. I must receive your solutions at the end of the class on the day the problem set is due.

When computing your problem set average for grading purposes, I will drop the lowest score, that is only your best $n - 1$ problem sets will be taken into consideration. Therefore, you can miss a problem set without repercussions.

Exams

You will have three exams, two midterms and one final exam. Each midterm is non-cumulative. We will hold the midterms in the usual lecture room. The dates for the midterms will be announced in class and in BlackBoard. If for any reason you anticipate that you won't be able to take the exam on that date, please let me know as soon as possible. The final exam is cumulative and will cover all topics covered in the course. The final will be given during finals week, on Tuesday, May 14 2019 from 5:30PM to 8:00PM, according to the Registrar's schedule.

Topics we will be covering

1. Introduction and motivation; the market–demand and supply, equilibrium, comparative statics, market structure, Pareto Efficiency, difference between positive and normative analysis.
2. Consumer choice: budget sets.
3. Consumer choice: preferences and utility.
4. Consumer choice: Demand
5. Choice under uncertainty and inter temporal choice.
6. Market Demand, Consumer surplus and elasticity.
7. Market Equilibrium.
8. The firm: Technology.

9. The firm: Profit maximization.
10. The firm: Cost minimization.
11. Firm and Industry Supply.
12. Monopoly.
13. Externalities and Public Goods.
14. Game Theory and Oligopoly.
15. Exchange and Production in general equilibrium.
16. Depending on time, we may cover some of the following:
 - Asset markets and risky assets.
 - Asymmetric information.
 - Auctions and Information Technology.

Student Accessibility Support Center Statement

If you have a physical, psychological, medical or learning disability that may impact your course work, please contact Student Accessibility Support Center, ECC (Educational Communications Center) Building, Room 128, (631)632-6748. They will determine with you what accommodations, if any, are necessary and appropriate. All information and documentation is confidential.

Students who require assistance during emergency evacuation are encouraged to discuss their needs with their professors and Student Accessibility Support Center. For procedures and information go to the following website: <http://www.stonybrook.edu/ehs/fire/disabilities>.

Academic Integrity

Each student must pursue his or her academic goals honestly and be personally accountable for all submitted work. Representing another person's work as your own is always wrong. Faculty is required to report any suspected instances of academic dishonesty to the Academic Judiciary. Faculty in the Health Sciences Center (School of Health Technology & Management, Nursing, Social Welfare, Dental Medicine) and School of Medicine are required to follow their school-specific procedures. For more comprehensive information on academic integrity, including categories of academic dishonesty please refer to the academic judiciary website at http://www.stonybrook.edu/commcms/academic_integrity/index.html

Critical Incident Management

Stony Brook University expects students to respect the rights, privileges, and property of other people. Faculty are required to report to the Office of University Community Standards any disruptive behavior that interrupts their ability to teach, compromises the safety of the learning environment, or inhibits students' ability to learn. Faculty in the HSC Schools and the School of Medicine are required to follow their school-specific procedures. Further information about most academic matters can be found in the Undergraduate Bulletin, the Undergraduate Class Schedule, and the Faculty-Employee Handbook.

Disclaimer

Subsequent changes can be made to any part of this syllabus when necessary. Any changes will be announced in class and on BlackBoard.